Lesson 2: Electrical Circuits – What Causes That Light Bulb Moment?

GCSE Physics (Gateway Science) – Core Practical Focus



Teacher: Greg

Gemini Notebook

She’s electric,
she has a family full
of eccentrics...



You already know about Static Electricity (charge separation trapped on insulators via friction).

But what happens when we set those charges free? What actually causes that light bulb moment when a continuous flow of electrons moves through a closed conducting loop?

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Define static electricity.

Build-up and separation of electrical charge carried by electrons, caused by friction.

What force causes hair to stand on end rubbed with a balloon?

Electrostatic repulsion.

How do you calculate the Rf value of a chromatography spot?

Distance moved by spot ÷ distance moved by solvent.



What are the main parts of natural selection?

Variation, Selection pressure, Survival, Inheritance.

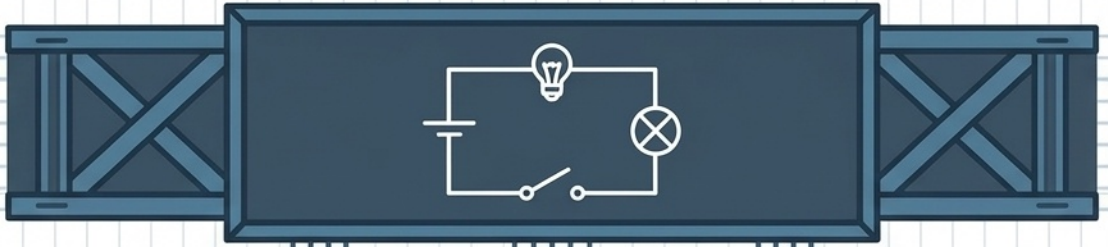
Draw or describe metallic bonding.

Regular lattice of positive metal ions surrounded by a sea of delocalised electrons.

What causes the 'light bulb moment' in a circuit?

Potential difference from a power source pushes delocalised electrons through the filament wire against resistance.

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Define

Master the invisible forces: Potential Difference, Electric Current, and Electrical Resistance.

Construct

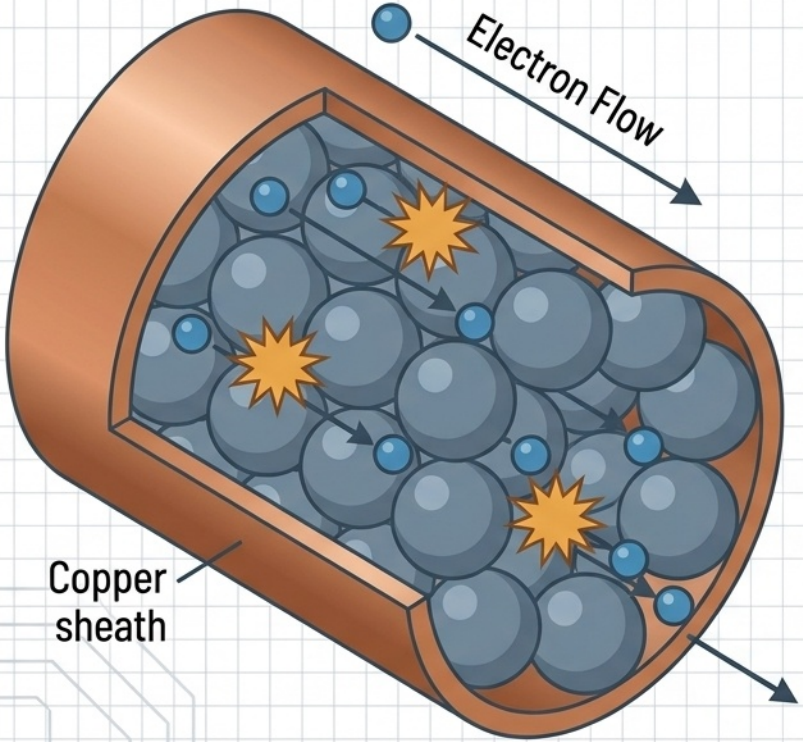
Build simple DC circuits to accurately measure potential difference and current.

Challenge

Understand and apply the Ohm's law equation ($V = I \times R$).

Keyword Toolkit: Electricity, Current (Amps), Potential difference / Voltage (Volts), Resistance (Ohms).

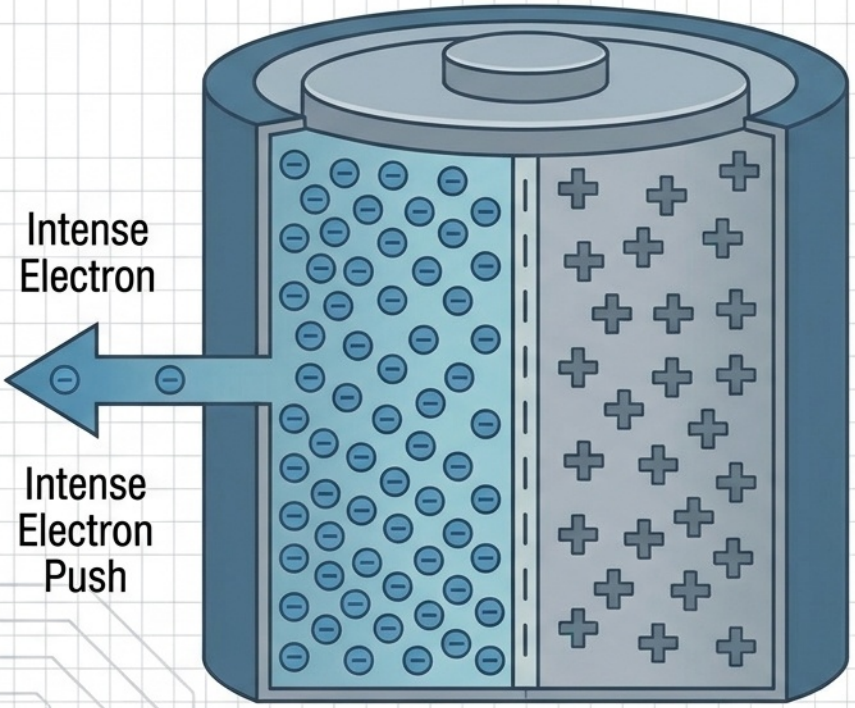
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The Microscopic Mechanism of Conduction

- The Drift: Delocalised electrons in a metallic conductor drift from higher electrical potential (negative terminal) to lower electrical potential (positive terminal).
- The Light Bulb Moment Explained: Potential difference pushes these electrons through the thin filament wire. They collide with the vibrating positive metal ions in the conductor lattice. This resistance dissipates electrical energy into radiant light and thermal energy.

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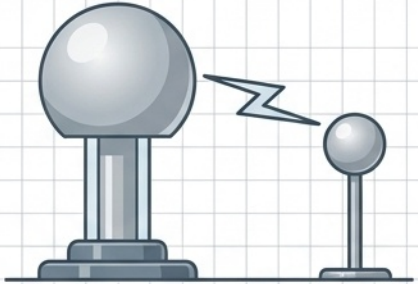
The Engine: How a Battery Works

Step 1: Chemical Separation: Internal chemical reactions separate charges.
Step 2: Potential Difference: This separation establishes an electrical potential difference—a severe imbalance in electron density.

Step 3: Continuous Flow: When a closed circuit is connected, intense electron repulsion at the negative terminal pushes electrons continuously through the entire circuit loop.

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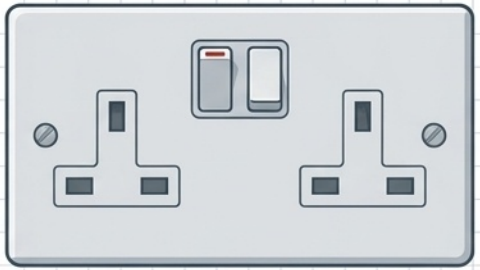
The Great Paradox: “Current Kills”



Van de Graaff Generator

Voltage: 50,000 V. Massive energy per charge.
Current: Microscopic rate of flow.
Result: Safe static shock.

Mains Electricity



Voltage: 230 V. Lower energy per charge.
Current: Massive, sustained rate of flow.
Result: Lethal continuous current.

Synthesis: Voltage determines how much energy each electron carries. Current determines how many electrons are hitting you per second. Rate of flow is what makes electricity dangerous.

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Electric Current (I)

Rate of flow of electrical charge (electrons passing a point per second).

Formula: $Q = I \times t$

Unit: Amperes / Amps (A)

Metaphor: River flow (a placid brook vs. a raging torrent).



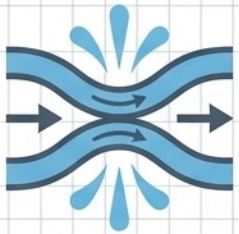
Potential Difference (V)

Measure of energy transferred per unit charge.

Formula: $V = \frac{E}{Q}$

Unit: Volts (V)

Metaphor: Waterfall height (a higher vertical drop transfers greater gravitational potential energy).



Electrical Resistance (R)

Opposition to flow; electrons colliding with the positive metal ion lattice. No normal component is resistance-free.

Formula: $V = I \times R$ (Ohm's Law)

Unit: Ohms (Ω)

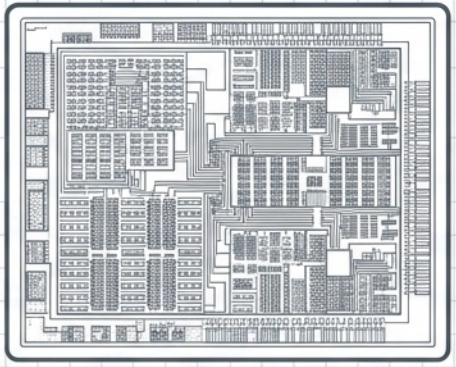
Metaphor: A pinched hosepipe or a rocky riverbed restricting flow.

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The Universal Language of Circuits

Cell Rate of cell	**Battery** Multiple cells	**Open Switch** Open switch	**Closed Switch** Closed switch	**Lamp** Laap function	**Ammeter** Current charge
Voltmeter Measur energy per charge	**Fixed Resistor** Resittor resistor	**Variable Resistor** Pro variable resistor	**Variable Resistor** Fiof variable resistor	**Thermistor** Areas hockey stick line	**LDR** Por rectangle in printing
Diode Triangle diode	**LED** Diode anmals points out	**Fuse** Einronged fuse			

Why use symbols?: They provide absolute clarity, universality, and speed in circuit design.



Apple Silicon Microchip Cross-Section

The Scale of Design: Modern microchips, like Apple Silicon, pack over 2 million transistors into a cross-section the width of a human hair. We use the exact same schematic logic to design them.

Core Symbols to Master: Cell, battery, open/closed switch, lamp/bulb, ammeter, voltmeter, fixed/variable resistors, thermistor, LDR, diode, LED, fuse.

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60-Second Challenge: Symbol Snap

Instructions:
Pair-and-share.
Match the physical component to its universal schematic symbol before the timer runs out.

Physical Components







Abstract Symbols



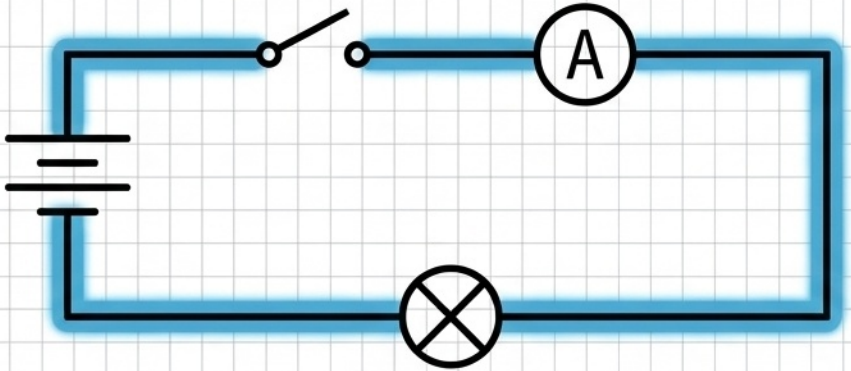




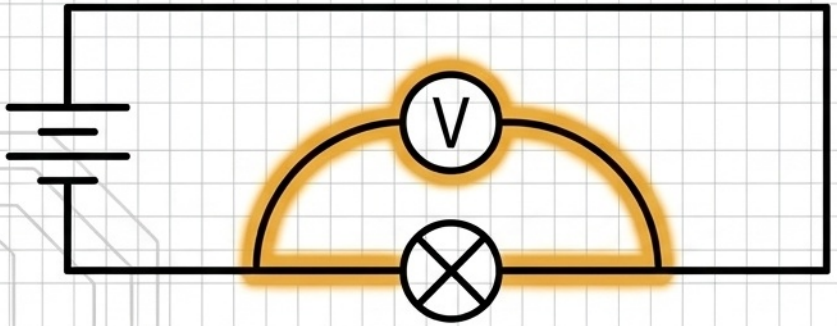


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The Golden Rules of Measurement



Measuring Flow (Ammeter):
Because it measures the rate of flow of charge, an Ammeter must be connected strictly in SERIES connected strictly in SERIES with the component.



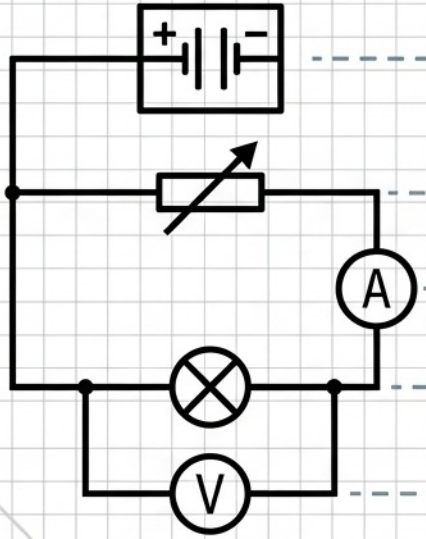
Measuring Energy (Voltmeter):
Because it measures the potential difference (energy drop) across a component, a Voltmeter must be connected strictly in PARALLEL across the component.

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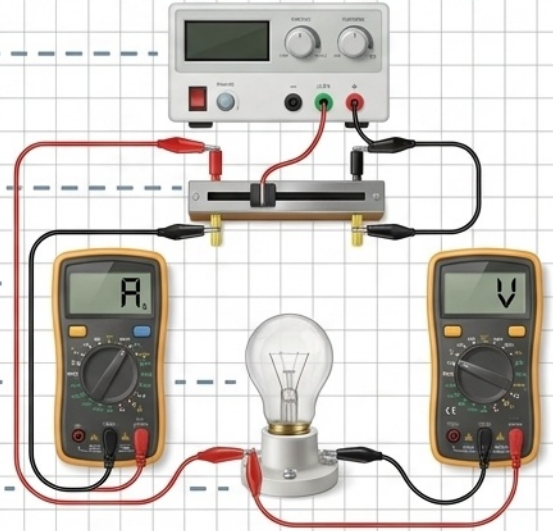
Core Practical: Filament Lamp Characteristics

Objective: Construct a DC circuit to investigate the I-V characteristics of a filament lamp.

Abstract Symbols



Physical Components



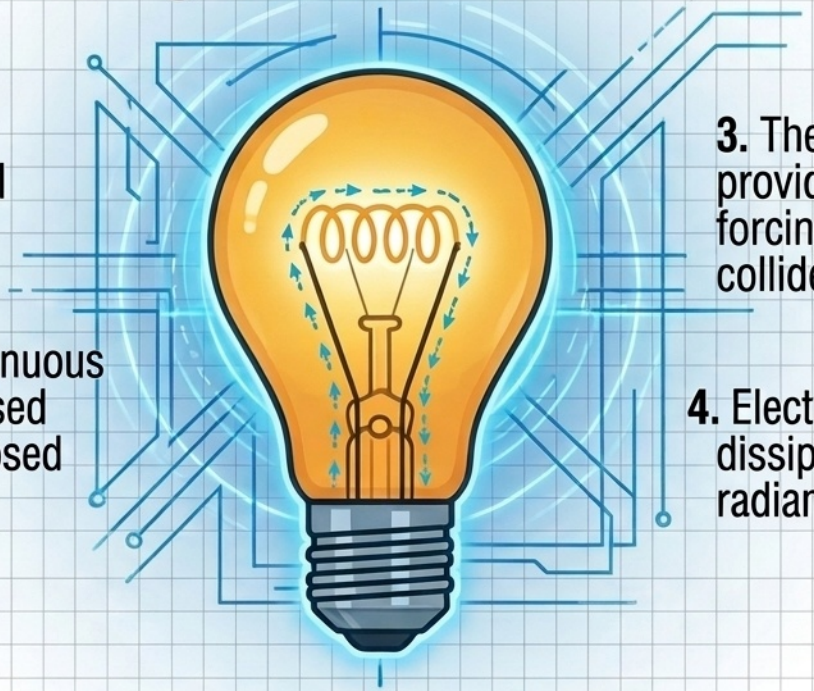
Key Action: Adjust the variable resistor to alter the potential difference and observe the resulting change in current.

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The Anatomy of a Light Bulb Moment

1. A chemical engine establishes a Potential Difference (V).

2. This pushes a continuous Current (I) of delocalised electrons around a closed loop.



3. The filament lattice provides Resistance (R), forcing electrons to collide with metal ions.

4. Electrical energy is dissipated as heat and radiant light.

The Rule: You now speak the language of the electrons. Time to build your circuits.

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